

Targeted EDMD Completeness Test for the Three-Mode Oscillatory Mode

Analysis-only numerical audit

7 September 2026

Abstract

The predeclared raw-action/cross-term extension exposes a complex EDMD candidate with imaginary part near the independently measured frequency 5.1–5.5. Three of four bath/timestep cases pass every frozen candidate gate, but the driven $\Delta t = 10^{-3}$ case fails dictionary convergence: at $\tau = 0.05$, E1, E2 and E3 give real parts $-6.466, -5.112, -5.520$, whose range 1.354 exceeds the frozen tolerance 1.104. Consequently the requested four-case completeness test *fails*; the EDMD spectrum remains incomplete. The extension also moves the slowest visible real rate from approximately -0.95 to approximately -0.90 – -0.92 (non-overlapping old/new confidence intervals in three of four cases). Thus the old -0.95 value was dictionary-sensitive and cannot be promoted to the slowest rate of the full generator spectrum.

1 Frozen question and interpretation gate

This is not a new stochastic simulation. It reuses the four controlled $n = 3$ trajectory ensembles (64 independent streams, burn-in 500, stationary time 1000, snapshots every 0.01) from the double-precision, fixed-timestep, standard-trigonometry, projection-free implementation. The protocol was frozen in commit 0af6652 before any extended spectrum was computed.

The original D1/D2/D3 log-action–Fourier bases were augmented once with the same 19 predeclared raw features: I_1, I_2, I_3 , their six quadratic products, and the ten requested functions built from $dI = I_1 - I_3$, including $dI \cos \theta_1, dI \sin \theta_1, dI \cos \theta_3, dI \sin \theta_3, dI \cos(\theta_1 - \theta_3)$ and $dI \sin(\theta_1 - \theta_3)$. The nominal sizes are E1/E2/E3= 109/269/519. No function was added after looking at the spectra.

The frozen interpretation was: recover the oscillation and leave the leading real confidence interval unchanged to promote the slow rate; recover it but move the rate to diagnose the old rate as dictionary-limited; otherwise declare the spectrum incomplete. The last branch applies.

2 Primary result: targeted oscillatory mode

The reference in each case was chosen deterministically from E3 at $\tau = 0.05$: $4.5 \leq \Im \lambda \leq 6$, $-20 \leq \Re \lambda \leq -0.05$, then largest real part. Bootstrap intervals resample the 64 complete trajectories 500 times.

Table 1: Primary complex candidate. “AC” is the independent damped- oscillation estimate from $I_1 - I_3$. Intervals are trajectory-bootstrap 95% intervals.

case	$\Re \lambda$	95% CI	$\Im \lambda$	95% CI	AC $\Im \lambda$	accepted	gate
driven, 10^{-3}	-5.520	$[-6.027, -4.885]$	5.037	$[4.467, 5.674]$	5.401	500/500	FAIL
driven, $2.5 \cdot 10^{-4}$	-5.582	$[-6.132, -4.828]$	5.696	$[5.077, 6.302]$	5.500	500/500	PASS
equal, 10^{-3}	-5.060	$[-5.877, -4.750]$	5.831	$[5.112, 6.339]$	5.081	500/500	PASS
equal, $2.5 \cdot 10^{-4}$	-5.963	$[-6.297, -5.403]$	5.293	$[4.778, 5.776]$	5.264	500/500	PASS

The independent frequency lies inside three EDMD intervals. In the equal 10^{-3} case it lies just below the EDMD lower endpoint (by 0.031), while the point estimates differ by 0.750. Therefore even the frequency comparison is not a four-case agreement. The EDMD real parts agree with the independently fitted driven damping rates $(-5.024, -5.003)$ within their broad EDMD intervals, but not with the equal-temperature damping rates $(-4.141, -4.184)$. This is additional evidence that the selected equal-temperature candidates should not be identified as the complete known mode without qualification.

2.1 Dictionary gate

Table 2: Matched candidates at $\tau = 0.05$, shown as $\Re\lambda + i\Im\lambda$. Only the first row violates the predeclared 20% stability rule.

case	E1	E2	E3	gate
driven, 10^{-3}	$-6.466 + 4.904i$	$-5.112 + 4.918i$	$-5.520 + 5.037i$	FAIL
driven, $2.5 \cdot 10^{-4}$	$-5.404 + 6.194i$	$-5.698 + 5.066i$	$-5.582 + 5.696i$	PASS
equal, 10^{-3}	$-5.010 + 5.464i$	$-5.768 + 5.315i$	$-5.060 + 5.831i$	PASS
equal, $2.5 \cdot 10^{-4}$	$-5.476 + 5.692i$	$-5.736 + 5.416i$	$-5.963 + 5.293i$	PASS

For the failed row, the real-part range is 1.3536 versus tolerance 1.1039; the imaginary-part range is 0.1329 versus tolerance 0.9836. Thus it is the damping estimate, not the frequency, that fails dictionary convergence. Retained/nominal Gram dimensions (E1,E2,E3) were respectively 107/109, 267/269, 517/519; 106/109, 266/269, 516/519; 107/109, 267/269, 516/519; and 107/109, 267/269, 517/519 in table order.

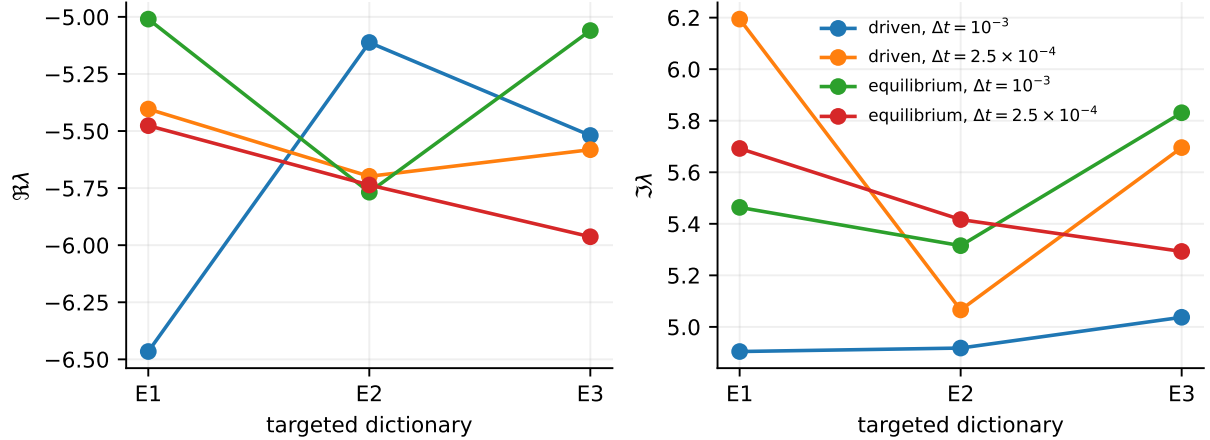


Figure 1: Dictionary dependence at the predeclared reference lag.

2.2 Short- and long-lag checks

All three short lags are far below their principal-log Nyquist bounds and pass the frozen lag-spread gate. The exact matched values are:

Table 3: E3 short-lag candidates. Nyquist bounds for $\tau = 0.02, 0.05, 0.10$ are 157.080, 62.832 and 31.416, respectively; none is aliased.

case	$\tau = .02$	$\tau = .05$	$\tau = .10$
driven, 10^{-3}	$-5.673 + 5.271i$	$-5.520 + 5.037i$	$-5.307 + 5.305i$
driven, $2.5 \cdot 10^{-4}$	$-5.194 + 5.709i$	$-5.582 + 5.696i$	$-5.296 + 5.409i$
equal, 10^{-3}	$-5.252 + 5.609i$	$-5.060 + 5.831i$	$-4.410 + 6.500i$
equal, $2.5 \cdot 10^{-4}$	$-5.602 + 5.294i$	$-5.963 + 5.293i$	$-5.913 + 5.008i$

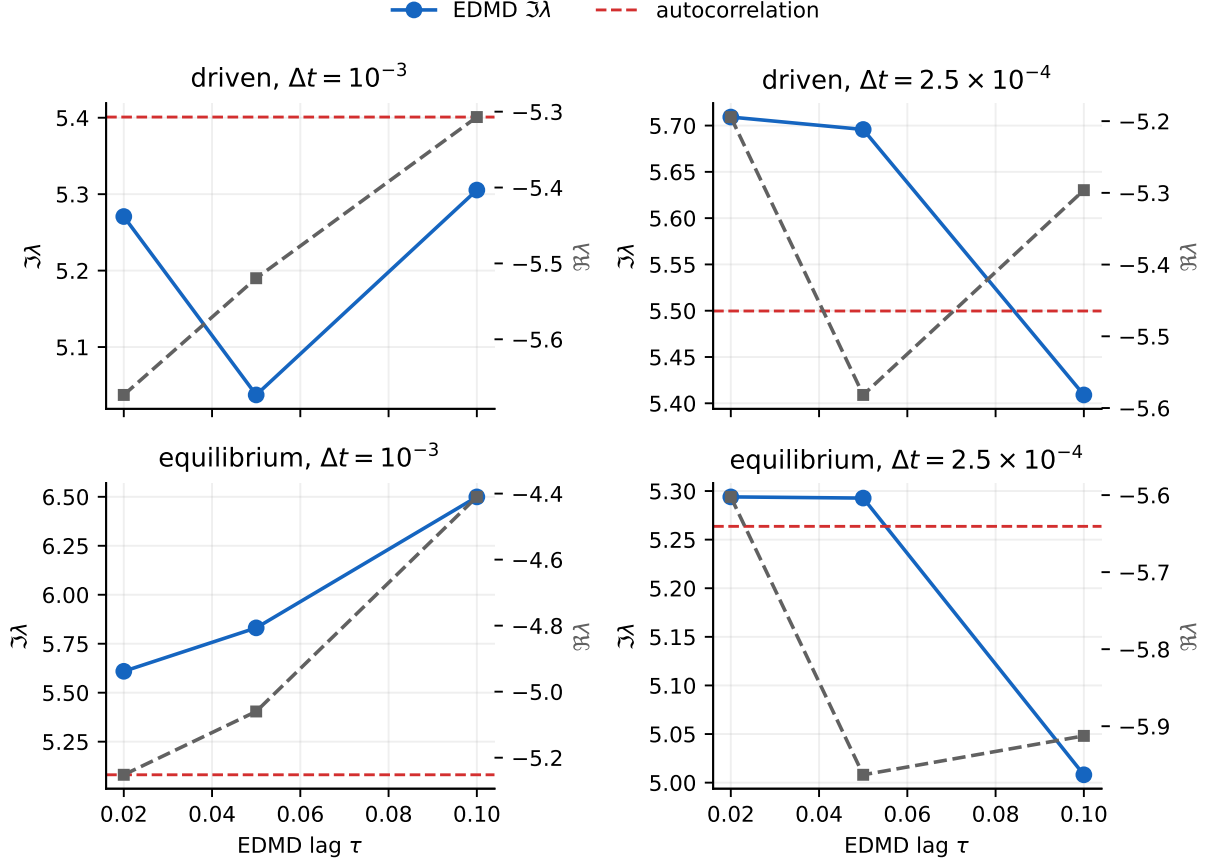


Figure 2: Short-lag convergence. Blue is the matched EDMD frequency; red is the independently fitted $I_1 - I_3$ frequency. Grey, on the right axes, is the EDMD real part.

Long lags provide the predicted aliasing diagnostic rather than evidence for convergence. At $\tau = 0.25$ (Nyquist 12.566), no row is aliased. At $\tau = 0.50$ (Nyquist 6.283; 20% exclusion threshold 5.027), three of four rows are aliased; only driven 10^{-3} , with $\Im \lambda = 4.951$, remains just outside the exclusion region. At $\tau = 1$ all four matches sit exactly at $\Im \lambda = \pi$ and are marked aliased. The long-lag values are retained in `oscillatory_convergence.csv`; they do not enter the short-lag gate.

The cutoff check is numerically flat to the printed precision at $10^{-8}, 10^{-10}, 10^{-12}$ in every case. At $E3/\tau = .05$, the primary Gram condition numbers range from $6.41 \cdot 10^8$ to $6.38 \cdot 10^9$; the truncated 10^{-8} fits retain 515 directions and have condition numbers $2.19 \cdot 10^5$ – $3.25 \cdot 10^5$, yet produce the same target eigenvalues. Every constant-mode error is below $1.28 \cdot 10^{-12}$ (required 10^{-8}).

3 Leading visible real mode

The targeted extension changes the leading visible real mode. Each new row passes dictionary, original-lag, timestep, cutoff, 500-bootstrap and neighbour separation gates, but “reportable” here means only within the tested EDMD subspaces, not proven leading in the full spectrum.

Table 4: Original and enlarged-dictionary leading visible real rates.

case	old	old 95% CI	new	new 95% CI	disjoint?
driven, 10^{-3}	-0.9402	$[-0.9546, -0.9208]$	-0.8998	$[-0.9164, -0.8826]$	yes
driven, $2.5 \cdot 10^{-4}$	-0.9632	$[-0.9800, -0.9419]$	-0.9224	$[-0.9435, -0.9024]$	no (marginal)
equal, 10^{-3}	-0.9551	$[-0.9724, -0.9328]$	-0.9058	$[-0.9227, -0.8886]$	yes
equal, $2.5 \cdot 10^{-4}$	-0.9408	$[-0.9548, -0.9230]$	-0.8945	$[-0.9118, -0.8777]$	yes

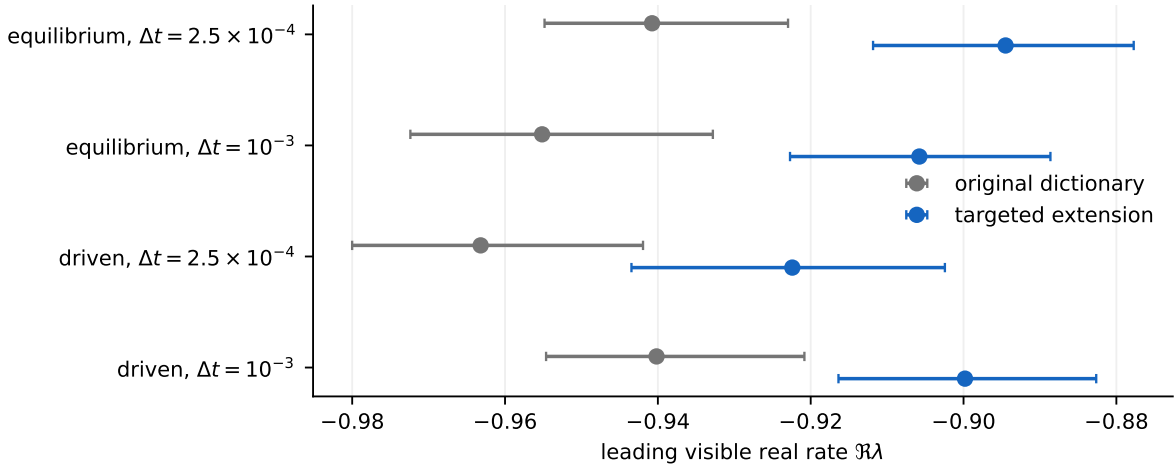


Figure 3: The extension shifts all four leading visible point estimates toward zero. Three old/new interval pairs are disjoint.

The two driven timesteps now give -0.9003 and -0.9227 ; the two equilibrium timesteps give -0.9055 and -0.8942 . The rate is therefore stable near -0.91 within the enlarged dictionary, but the movement establishes that the old -0.95 estimate was dictionary-limited. Because the oscillatory completeness gate fails, -0.91 is still only the slowest mode visible to E1–E3.

The two driven candidates near -1.8 and -2.1 do not separate. Their bootstrap intervals are $[-1.900, -1.558]$ and $[-2.159, -1.786]$ at $\Delta t = 10^{-3}$, and $[-1.882, -1.677]$ and $[-2.160, -1.789]$ at $\Delta t = 2.5 \cdot 10^{-4}$; both pairs overlap. This secondary test fails.

4 Updated modal weights

Table 5: Absolute covariance weights on the new leading visible mode.

case	I_2	$I_1 + I_3$	$\cos \theta_3$	cosine sum
driven, 10^{-3}	.492	.294	.121	.214
driven, $2.5 \cdot 10^{-4}$	.485	.285	.122	.218
equal, 10^{-3}	.470	.268	.103	.203
equal, $2.5 \cdot 10^{-4}$	.466	.270	.101	.202

The robust ordering $I_2 > I_1 + I_3$ survives and remains consistent with their controlled autocorrelation plateaus -0.935 and -1.003 . The strict monotone correspondence does *not* survive for the two cosine observables: the cosine-sum weight is larger than the $\cos \theta_3$ weight, while their plateaus are -1.122 and -1.092 , respectively. Modal weight alone therefore does not totally order the finite-window autocorrelation fits.

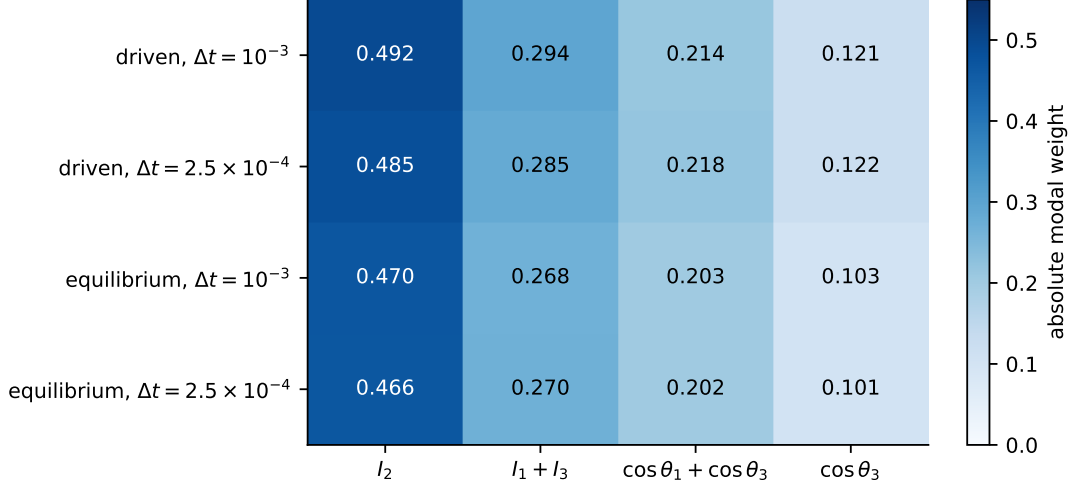


Figure 4: Leading-mode covariance weights after the dictionary extension.

5 Eigenfunction slices

The requested equal-action slices were computed at $I_1 = I_2 = I_3 = 1, 2, 4$ for both the selected oscillatory candidate and the leading visible real mode in all four cases. Figure 5 shows one representative case. Since the overall oscillatory gate fails, the upper row is a diagnostic candidate eigenfunction, not a validated generator eigenfunction. All four raw slice tables and figures are included in the package.

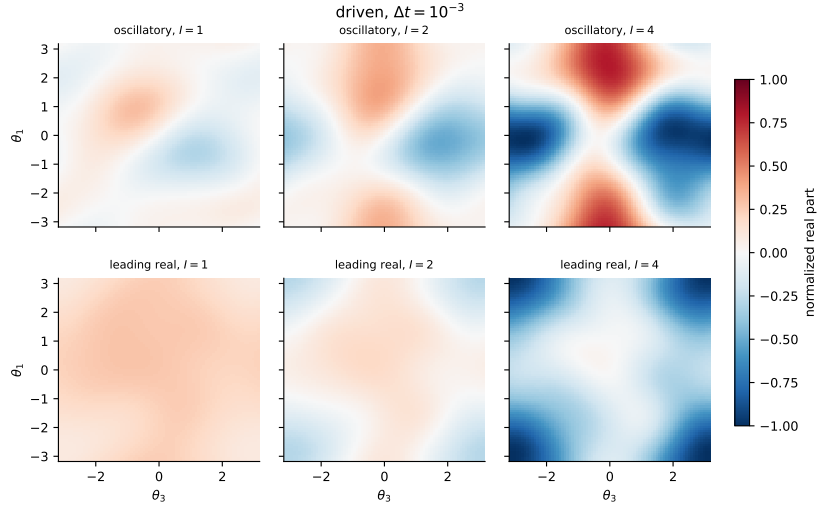


Figure 5: Normalized real parts of the two EDMD eigenfunctions for the driven $\Delta t = 10^{-3}$ case.

6 Provenance and reproducibility

The analysis source was frozen in commit 70cbab9; its SHA-256 is a986b0cc1d3f0e190223d348d7522b2ac81ce307e0628fd00852b2329aee2c57. The imported original EDMD helper has SHA-256 d4c53270ed902cbcf6692dcc64d7c1f423e40189504d03703c6e4f8cd50cc6ef. The controlled trajectory simulator source and binary hashes are respectively 58c871882f1180701a7aae637a8f4be113315a72fb9d7815562f5fc90faf7976 and f7cd820bb8f716a8790f688db060259da5657e4f20fa09100546e66efb696b3f. Base stream seeds were 2026090801–2026090804 by case; each generated 64 independent stream seeds according to the saved controlled-run manifest. Bootstrap base seed was 2026090702, offset by case; real-mode bootstrap used an additional offset 100.

The exact analysis command was:

```
python3 experiments/edmd_oscillatory_extension_2026-09-07/  
run_extended_edmd.py \  
--input experiments/spectral_gap_controlled_2026-09-07/raw \  
--output experiments/edmd_oscillatory_extension_2026-09-07/analysis
```

followed by the frozen gate command:

```
python3 experiments/edmd_oscillatory_extension_2026-09-07/  
summarize_extended_edmd.py \  
--analysis experiments/edmd_oscillatory_extension_2026-09-07/analysis
```

The integrity audit found 256/256 input streams with the expected size, all four aggregate matrices with expected dimensions, 500/500 oscillatory bootstrap rows per case, 500 rows per tracked real mode, finite spectra, and all constant-mode checks passing. Full input and output SHA-256 manifests are in **provenance/**. The raw spectra, all per-replicate bootstrap fits, convergence tables, aggregate EDMD matrices, weights and eigenfunction slices are included rather than replaced by this report.

7 Claim boundary

The targeted extension substantially improves visibility of the known oscillatory frequency, but it does not pass the predeclared four-case dictionary-completeness test. The defensible statement is:

The enlarged raw-action EDMD dictionary reveals a reproducible complex candidate near the independently observed oscillation and a stable slow visible rate near -0.91 , but one dictionary gate fails and the slow rate moves relative to the original basis. Hence the complete generator spectrum and its spectral gap remain unresolved.

No additional basis was tested in this task.